Zachary Salloum

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Education

University of Michigan Ann Arbor, MI
Bachelor of Science in Mechanical Engineering *August 2025 – May 2029*

- Cumulative GPA: 3.96/4.00
- University of Michigan Regents Merit Scholarship Recipient, Fall 2025 and Winter 2026 Dean's Honor List

Professional Experience

Manufacturing and Operations Engineering Intern Charleston, SC
Bosch Charleston Plant *May 2026 – August 2026*

- Designed and machined alignment tooling to standardize technician procedure and reduce induction annealing machine rebuild downtime by 50%. Updated documentation to include alignment tool usage
- Reduced re-tray station stoppages by 40% by designing and machining custom loading fixtures for new work-piece carrier trays, ensuring proper tray alignment on the HDEV5 fuel injector chrome plating line
- Created comprehensive physical limit samples to increase shift standardization and reduce scrap by 10%. Audited operators to ensure compliance with scrap-handling procedures and limit samples
- Implemented a temporary pre-wash procedure for raw parts to combat contamination in the chrome line. This involved designing and fabricating custom wash trays and overseeing the washing of affected lots
- Used Microsoft PowerApps and Excel to create a streamlined app for shift leaders to log shift turnover data. Integrated this app with Microsoft PowerBI to easily track production and downtime metrics

Project Experience

Electric Outboard Motor Project | *UM Electric Boat Team* August 2025 – Present

- Designed a 50 HP electric propulsion system to meet our team's continuous output goal while satisfying the competition's battery voltage and total capacity restrictions of 55.5 V and 500 Ah
- Developed a unique, tri-motor drivetrain design to achieve output target while balancing competition constraints, available motor options, total outboard weight, and packaging requirements
- Engineered a custom 1.43:1 ratio gearbox, reducing motor speed by 30% and increasing torque output. Designed and machined brackets to mount motors, encoders, and inverters to the outboard power-head
- Created detailed parametric CAD assemblies in Siemens NX with attention to gearbox shaft alignment and manufacturability. Conducted FEA to validate gearbox structural integrity under expected loads

Custom-Built 3D Printer Project | *Passion Project* June 2023 – August 2024

- Independently designed and built an FDM 3D printer capable of 30% greater print speed and 50% larger total build volume than comparable consumer-grade printers, while maintaining excellent print quality
- Tested and tuned motion and extrusion systems to achieve a dimensional accuracy of ± 0.05 mm at print speeds of over 100mm/s. Iterated motor and linear rail mount designs to increase toolhead stability
- Designed and fabricated over 20 components in order to mount different build elements. Created an accurate parametric CAD assembly in Fusion 360 and generated a comprehensive bill of materials
- Integrated various electronics, including a display, main-board, power supply, and MOSFET. Modified main-board firmware to ensure compatibility with the custom hardware configuration

Skills

Software: SolidWorks, Siemens NX, Fusion 360, MATLAB, C++, Microsoft Office Suite, SAP, Power BI

Languages: English (native), French (spoken and written proficiency)

Relevant Coursework: Multivariable Calculus, Differential Equations, Physics I + II, Solid Mechanics